ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Gregor Johann Mendel	Mendel worked on garden pea plants.
2	**(c)** all round yellow	Round and yellow are dominant.
3	**(b)** the type of sperm that fertilises the egg	Father's sperm may carry X or Y.
4	**(c)** it has many contrasting characters and is easy to grow	Pea has clear contrasting traits and short life cycle.
5	**(b)** 9 : 3 : 3 : 1	Four phenotypes appear.
6	**(c)** XX	Both are X.
7	**(d)** heredity	Definition of heredity.
8	**(b)** 3	The 3 round green seeds are R_yy.
9	**(c)** XY	One X and one Y.
10	**(b)** all tall	Tallness is dominant.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heredity: transmission of characters from parents to offspring (1); Variation: differences among individuals of a species (1).	1+1
12	A cross involving one pair of contrasting characters (1); F2 ratio is 3 : 1 (1).	1+1
13	A trait gained during the lifetime of an individual (1); Example: loss of body weight due to starvation (1).	1+1
14	They have clear contrasting characters (1); They are easily grown and have a short life cycle (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	F1: genotype Tt, phenotype tall (1); F2 phenotypic ratio 3 tall : 1 dwarf (1); F2 genotypic ratio 1 TT : 2 Tt : 1 tt (1).	1+1+1
16	Eggs always carry an X chromosome (1); Sperm can carry X or Y (1); The sperm that fertilises the egg decides the sex (1).	1+1+1
17	DNA carries genetic information (1); Segments called genes control traits (1); DNA is passed from parents to offspring through gametes (1).	1+1+1
	OR: Round yellow x wrinkled green gives round yellow F1 (1); F2 shows round green and wrinkled yellow (1); New combinations show independent assortment (1).	1+1+1
18	Tall (T) is dominant over dwarf (t) (1); In Tt, tallness is expressed (1); Dwarfness appears only in tt (1).	1+1+1
	OR: F1 is the first filial generation from the cross of parents (1); F2 is produced by self-pollination of F1 (1); F2 shows the reappearance of recessive trait (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Parents: TT x tt (1); F1: all Tt, tall (1); F1 self-pollinated: gametes T and t (1); F2: TT, Tt, Tt, tt (1); Phenotypic ratio 3 : 1 and genotypic ratio 1 : 2 : 1 (1).	1+1+1+1+1
	OR: He crossed plants having contrasting traits (1); F1 showed only one parent's trait (1); F2 had both traits in 3 : 1 (1); Traits are controlled by pairs of factors (1); Factors segregate in gametes and are independently inherited (1).	1+1+1+1+1
20	Parents RRYY x rryy (1); F1 all RrYy, round yellow (1); F1 forms four gametes RY, Ry, rY, ry (1); F2 round yellow 9, round green 3, wrinkled yellow 3, wrinkled green 1 (1); Ratio 9 : 3 : 3 : 1 proves independent assortment (1).	1+1+1+1+1
	OR: Gametes: T, t and t, t (1); Offspring: Tt, Tt, tt, tt (1); Genotypic ratio 1 : 1 (1); Phenotypic ratio 1 tall : 1 dwarf (1); Such a cross is a test cross (1).	1+1+1+1+1

Mistakes to avoid:

- Giving 1 : 2 : 1 as the F2 *phenotypic* ratio of a monohybrid cross; it is 3 : 1 phenotypic and 1 : 2 : 1 genotypic.

- Saying that the dominant trait is 'stronger' or 'more common'; it simply masks the recessive allele in the heterozygote.
- Writing the genotype of F1 as TT or tt; the cross of pure TT and pure tt gives only Tt.
- Saying that the mother decides the sex of the child; all eggs carry X, and the father's sperm (X or Y) determines it.
- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.
- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** traits are inherited independently	Independent assortment.
2	**(c)** 46	23 pairs.
3	**(a)** 3 tall : 1 dwarf	Monohybrid ratio is 3 : 1.
4	**(d)** chromosomes	Chromosomes carry genes made of DNA.
5	**(c)** 44	22 pairs, i.e. 44.
6	**(b)** 1 : 2 : 1	TT : Tt : tt.
7	**(d)** genotype	Genotype is the genetic make-up.
8	**(b)** 50%	XY and XX are equally likely.
9	**(d)** dominant	It masks the other trait.
10	**(a)** heterozygous	Example: Tt.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Genes for different traits are inherited independently of each other (2).	2
12	They are not coded in the germ-cell DNA (1); Changes in body cells are not inherited (1).	1+1
13	Dominant trait expresses in the F1 (1); Recessive trait is hidden in F1 and appears in F2 (1).	1+1
14	23 in each (1); Zygote has 46 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inherited traits are controlled by genes; acquired are not (1); Inherited traits are passed on; acquired are not (1); Example: eye colour (inherited) and a scar (acquired) (1).	1+1+1
16	Autosomes: chromosomes that do not decide sex, 22 pairs (1); Sex chromosomes decide sex, 1 pair (1); Total 23 pairs (1).	1+1+1
17	Four types (1); RY, Ry, rY, ry (1); Each in equal proportion (1).	1+1+1
	OR: Each parent contributes one set of chromosomes (1); The child gets 23 from the father and 23 from the mother (1); Hence it has a mix of traits from both (1).	1+1+1
18	Small copying errors occur in DNA (1); These errors are passed to daughter cells (1); Variations accumulate over generations (1).	1+1+1
	OR: F1 has one dominant allele for each trait (1); Recessive alleles are masked (1); rryy forms only when both recessives combine in F2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dominant: expressed in F1, e.g. T (1); Recessive: hidden in F1, e.g. t (1); Homozygous: TT or tt (1); Heterozygous: Tt (1); Examples are from tall and dwarf pea (1).	1+1+1+1+1
	OR: Gametes carry different gene combinations (1); Fertilisation combines the genes of two parents (1); Copying errors add new variations (1); Variations help survival under changes (1); Variation is the basis of the diversity in nature (1).	1+1+1+1+1
20	Women have XX and men have XY (1); Egg always carries X (1); Sperm carries X or Y with equal probability (1); X sperm gives girl, Y sperm gives boy (1); The chance of a boy or a girl is 50% each (1).	1+1+1+1+1
	OR: One chromosome of each pair comes from each parent (1); Gametes have half the number (1); 23 in sperm, 23 in egg (1); Fertilisation restores 46 (1); So the number is constant (1).	1+1+1+1+1

Mistakes to avoid:

- Giving 1 : 2 : 1 as the F2 *phenotypic* ratio of a monohybrid cross; it is 3 : 1 phenotypic and 1 : 2 : 1 genotypic.

- Saying that the dominant trait is 'stronger' or 'more common'; it simply masks the recessive allele in the heterozygote.
- Writing the genotype of F1 as TT or tt; the cross of pure TT and pure tt gives only Tt.
- Saying that the mother decides the sex of the child; all eggs carry X, and the father's sperm (X or Y) determines it.
- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.
- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** not inherited	Acquired traits do not change DNA of germ cells.
2	**(c)** recessive	It is hidden in F1.
3	**(d)** TT or tt	Both alleles are the same.
4	**(a)** Eye colour	Eye colour is genetic.
5	**(d)** alleles	For example T and t.
6	**(b)** a test cross	It reveals the genotype of F1.
7	**(b)** variation within a species	Differences are inherited.
8	**(c)** phenotype	It is the observable character.
9	**(b)** T and t in equal numbers	Alleles separate during gamete formation.
10	**(c)** Sperm with X chromosome	X sperm gives XX.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	F1 plants carry both alleles (1); Alleles separate and combine in F2 (1).	1+1
12	Genotype: genetic make-up of an organism (1); Phenotype: its observable characteristics (1).	1+1
13	Egg carries X; sperm carries X or Y (1); X sperm gives a girl and Y sperm gives a boy (1).	1+1
14	A segment of DNA that codes for a protein (1); Present on chromosomes in the nucleus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	F1 is Tt (1); Gametes T and t combine at random (1); tt (dwarf) appears with probability 1/4 (1).	1+1+1
16	Gametes R, r and r, r (1); Genotypes Rr and rr in 1 : 1 (1); Half round, half wrinkled (1).	1+1+1
17	A Y-bearing sperm gives XY (1); XY is male (1); Absence of Y gives XX female (1).	1+1+1
	OR: Mother: XX produces only X eggs (1); Father: XY produces X and Y sperms (1); XX (girl) or XY (boy) in the ratio 1 : 1 (1).	1+1+1
18	Two parents contribute genes (1); Gametes combine in many ways (1); Crossing over and copying errors add to it (1).	1+1+1
	OR: Test cross: F1 crossed with a recessive parent (1); Self cross: F1 crossed with itself (1); Test cross reveals the genotype of F1 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Inherited: controlled by genes (1); Acquired: gained during lifetime (1); Examples: eye colour, hair colour (inherited) (1); Examples: scars, tanned skin (acquired) (1); Acquired changes do not alter germ-cell DNA (1).	1+1+1+1+1
	OR: Child gets half the genes from each parent (1); Traits are combined from both (1); Alleles may be recombined differently (1); Copying errors produce new variations (1); So the child has both resemblance and differences (1).	1+1+1+1+1
20	Genes are DNA segments that make proteins (1); Proteins control body characters (1); Gene for the plant hormone enzyme controls height (1); An efficient enzyme gives more hormone, so a tall plant (1); Less efficient enzyme gives dwarf plants (1).	1+1+1+1+1
	OR: DNA copying errors (1); Combination of gametes from two parents (1); Independent assortment of traits (1); Random fusion of gametes (1); Variation is passed to the next generation (1).	1+1+1+1+1

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- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

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